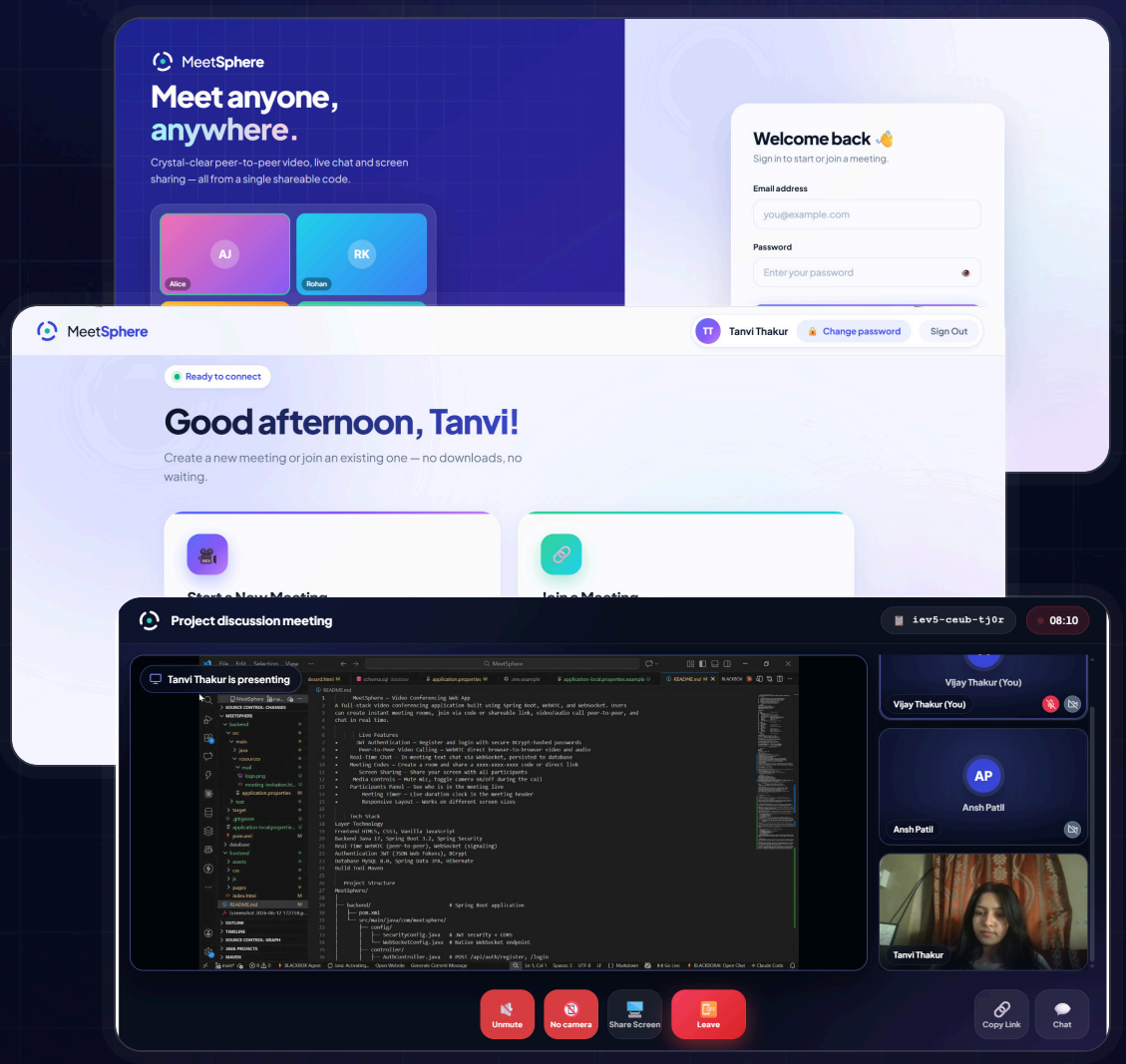
PROJECT PORTFOLIO

MeetSphere

Full-Stack Video Conferencing Application

A web application for online meetings, real-time video and audio communication, live chat, and screen sharing.

Java | Spring Boot | JavaScript | MySQL | WebRTC | WebSocket



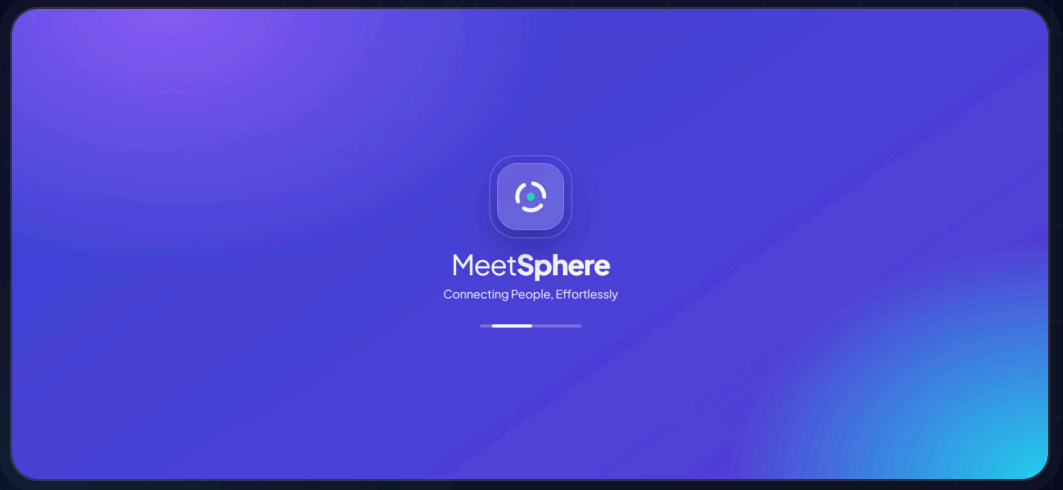
Personal Project | Tanvi Thakur | Java Full Stack Development

02 PROJECT OVERVIEW

What Is MeetSphere?

MeetSphere is a personal full-stack web application that explores the development of an online meeting platform. Users can create meeting rooms, share meeting codes, communicate through audio and video, exchange messages, and share their screens.

Personal Full-Stack Project | Runs Locally | Not Publicly Deployed



APPLICATION LAUNCH SCREEN

Account Management
User registration, sign-in, and password management.

Meeting Rooms
Create meetings and allow participants to join using a unique meeting code.

Video and Audio
Real-time peer-to-peer communication with camera and microphone controls.

Screen Sharing
Present your screen to other meeting participants.

Live Chat
Exchange messages during meetings.

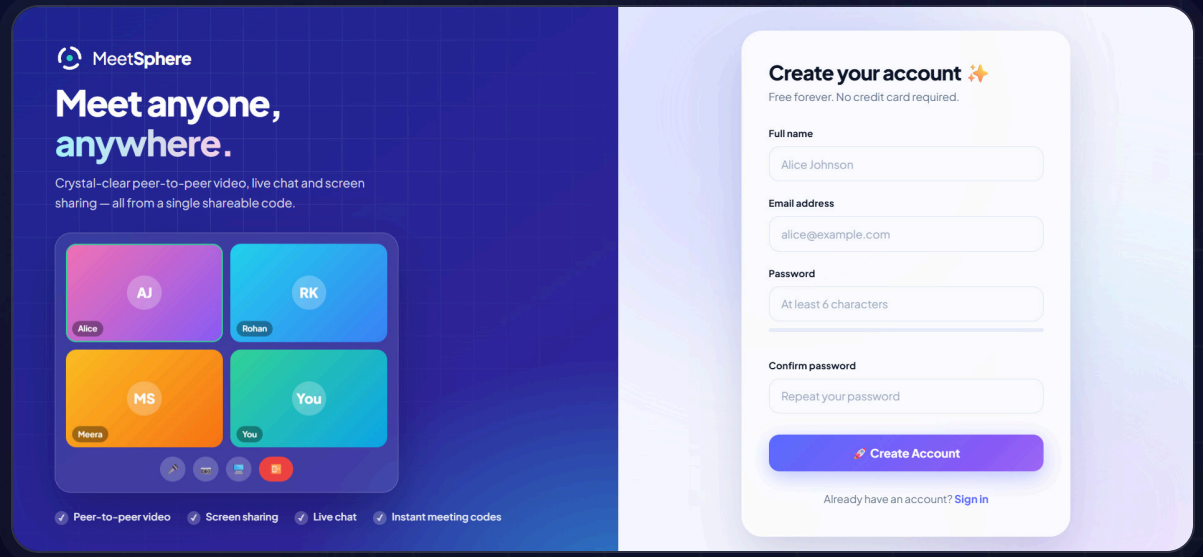
Participant Management
View meeting participants and their current camera and microphone status.

Meeting Timer
Display the meeting duration.

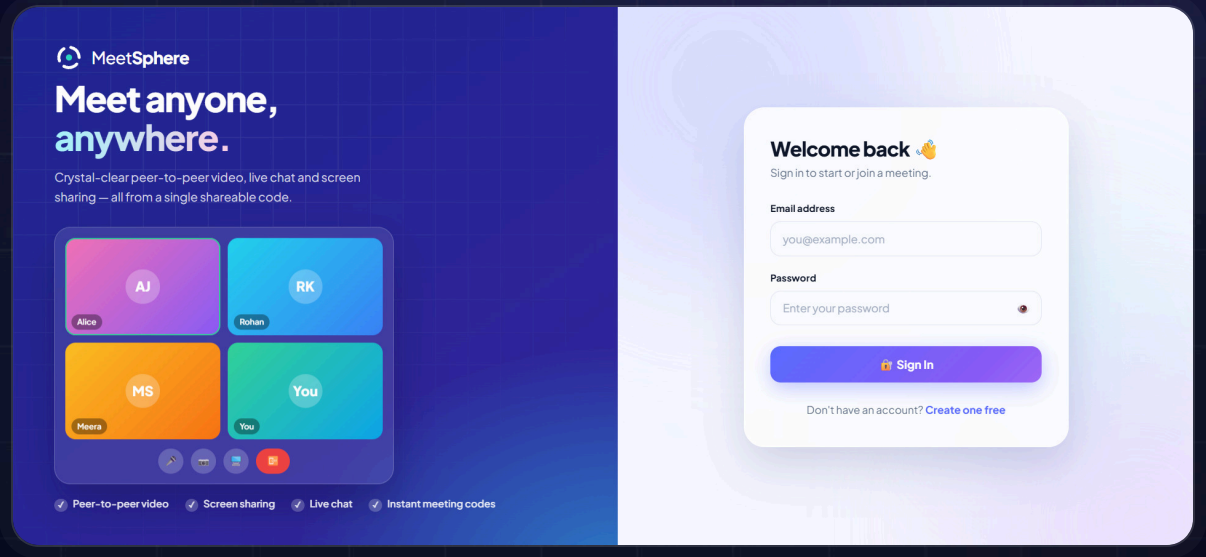
Email Invitations
Send meeting invitation emails containing meeting details and a joining link.

03 LOGIN & APPLICATION INTERFACE

Account Access



Sign-up. Users can create an account by entering their name, email address, password, and password confirmation. The interface includes password-strength guidance.



Sign-in. Users can sign in using their email address and password. The interface includes a password visibility toggle and an option to create an account.

CONSISTENT USER INTERFACE

A consistent visual style across the registration, sign-in, and application screens.

CORE APPLICATION FEATURES

Access to meeting creation, meeting codes, video communication, screen sharing, and live chat.

Start a Meeting | Join by Code | Account Controls

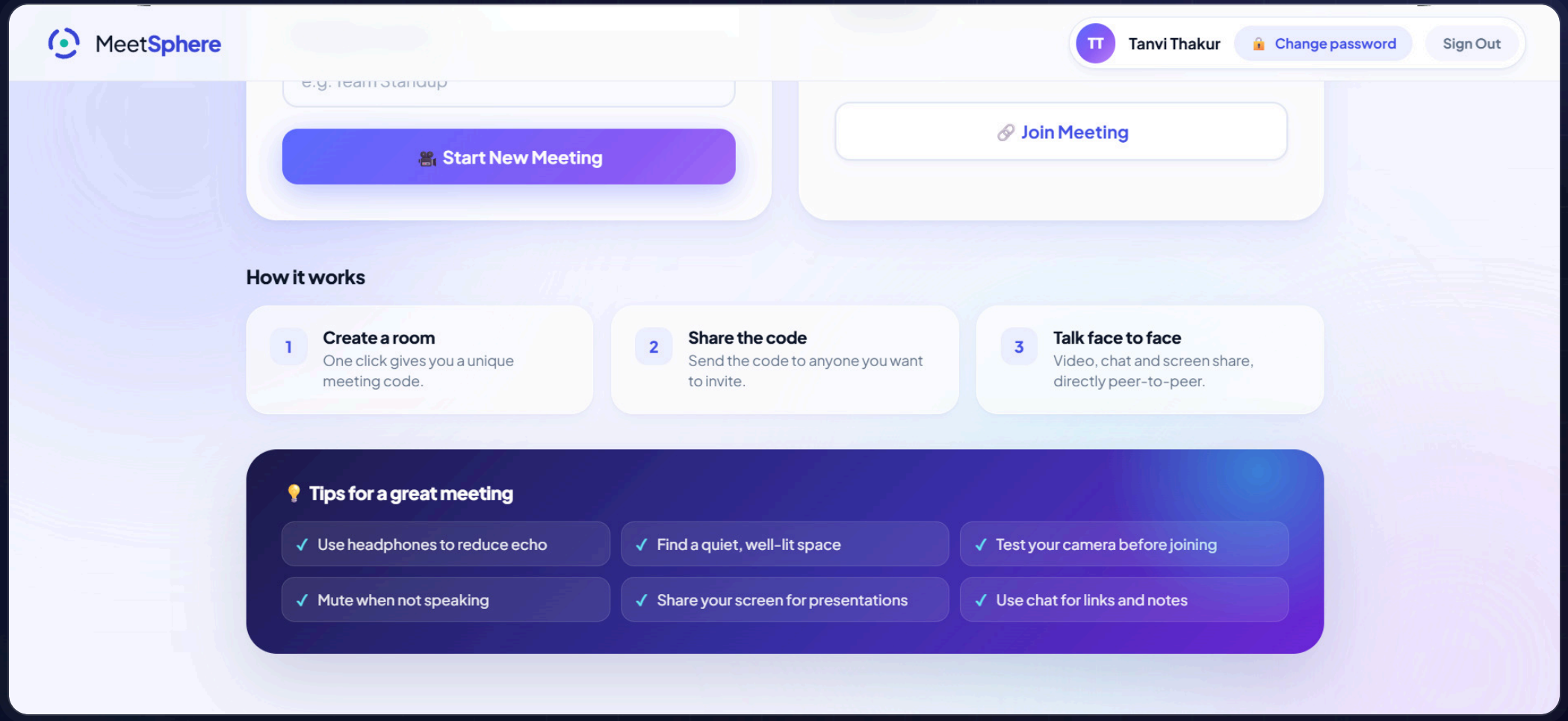
Dashboard After Sign-In



After signing in, users can start a new meeting or join an existing meeting using a meeting code. The dashboard also provides access to password management and sign-out functionality.

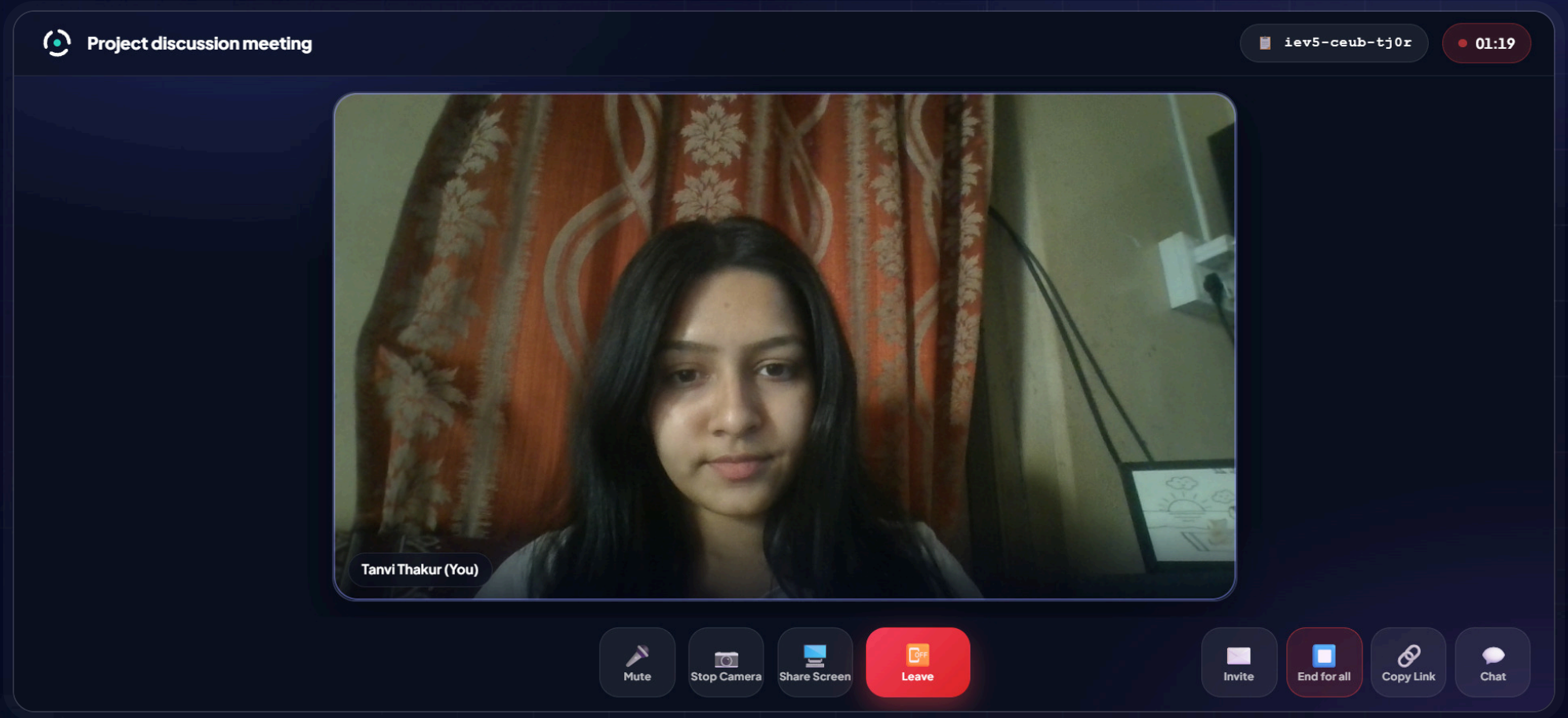
Create a Meeting | Share the Meeting Code | Join and Communicate

How It Works



MeetSphere provides a straightforward meeting workflow. Create a meeting room, share its meeting code with participants, and begin communicating through the meeting interface.

The Meeting Room



The meeting room brings the main communication features together in one interface. It displays the meeting title, meeting code, and meeting duration.

AVAILABLE CONTROLS: Mute | Camera | Share Screen | Leave Meeting | Invite Participants | End Meeting for All | Copy Link | Chat

Participant Tiles | Camera Status | Participant Panel

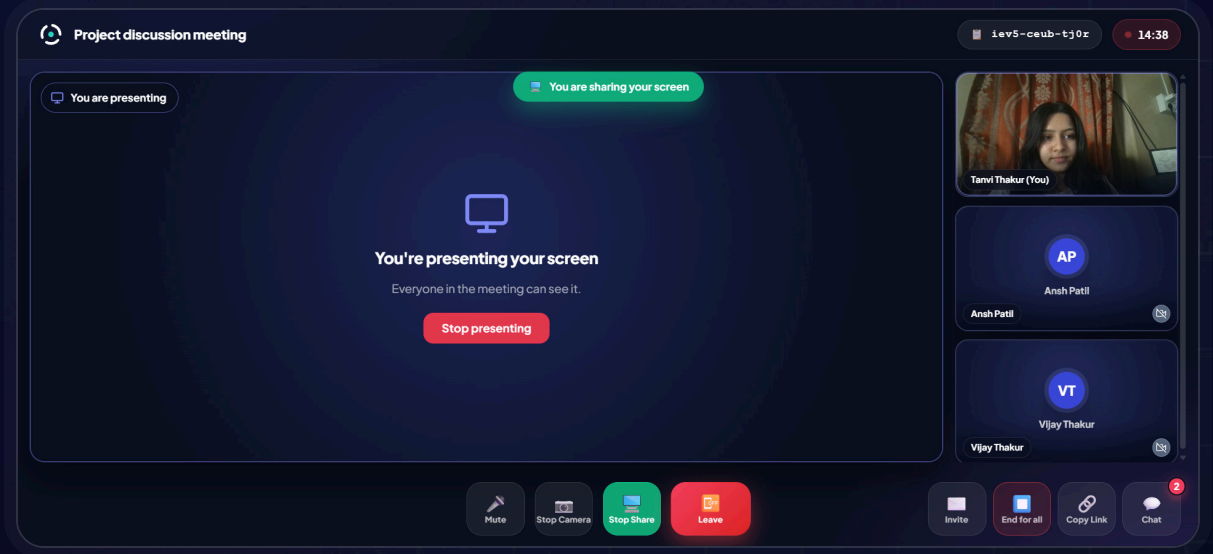
Participants and Camera States



The participant panel displays people connected to the meeting. Participants can appear in individual tiles, with camera-off states represented by an initials avatar and an appropriate status label.

05 SCREEN SHARING & MEETING EXPERIENCE

Presenting Your Screen





Screen sharing. Participants can share their screens during a meeting. The interface indicates when screen sharing is active and provides controls to stop sharing.

Participant panel. The People tab lists the participants in the meeting while the presentation continues.

PARTICIPANT PANEL

Shows who is in the meeting.

UNREAD CHAT INDICATOR

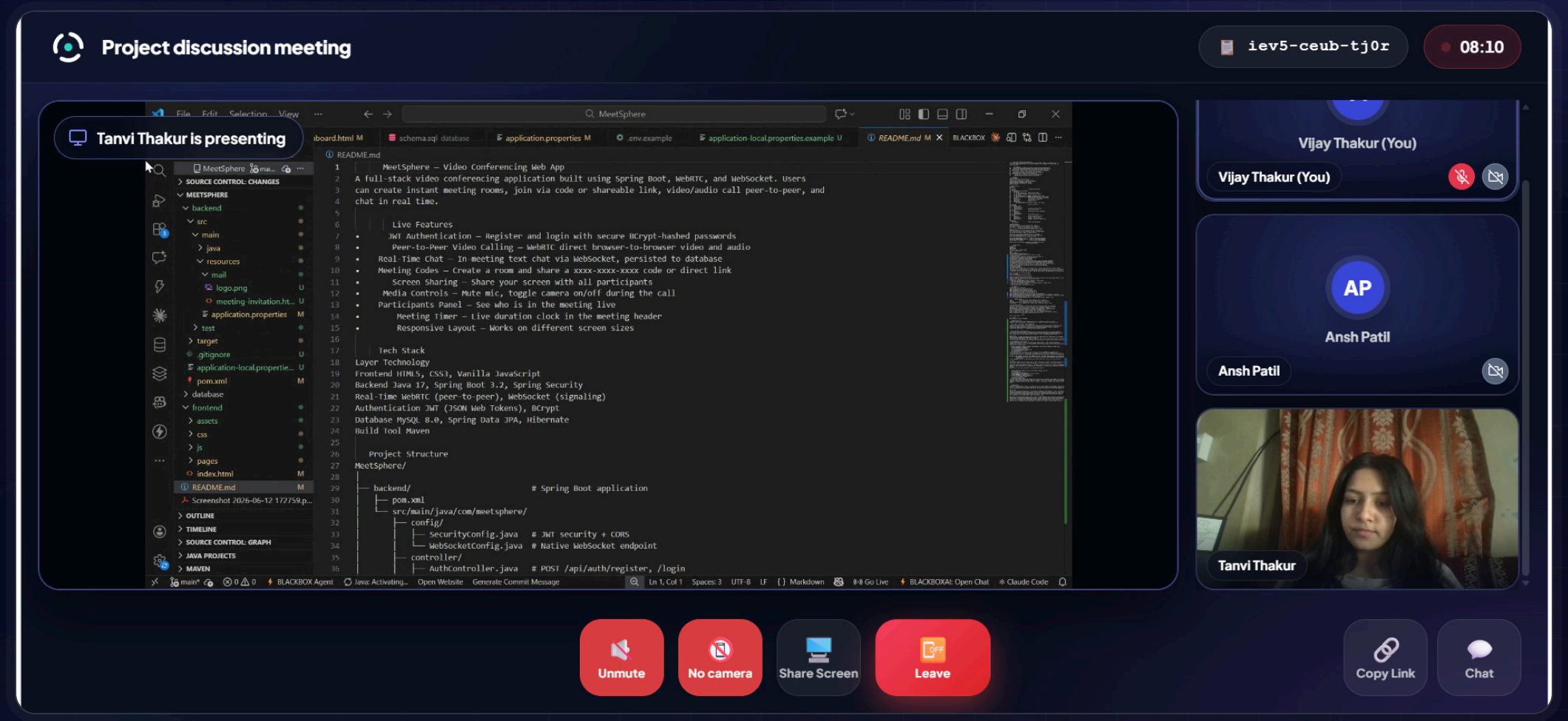
The Chat button shows a count of unread messages.

PRESENTATION LAYOUT

A presentation area with participant tiles alongside.

Shared Screen | Presenter Label | Participant Strip

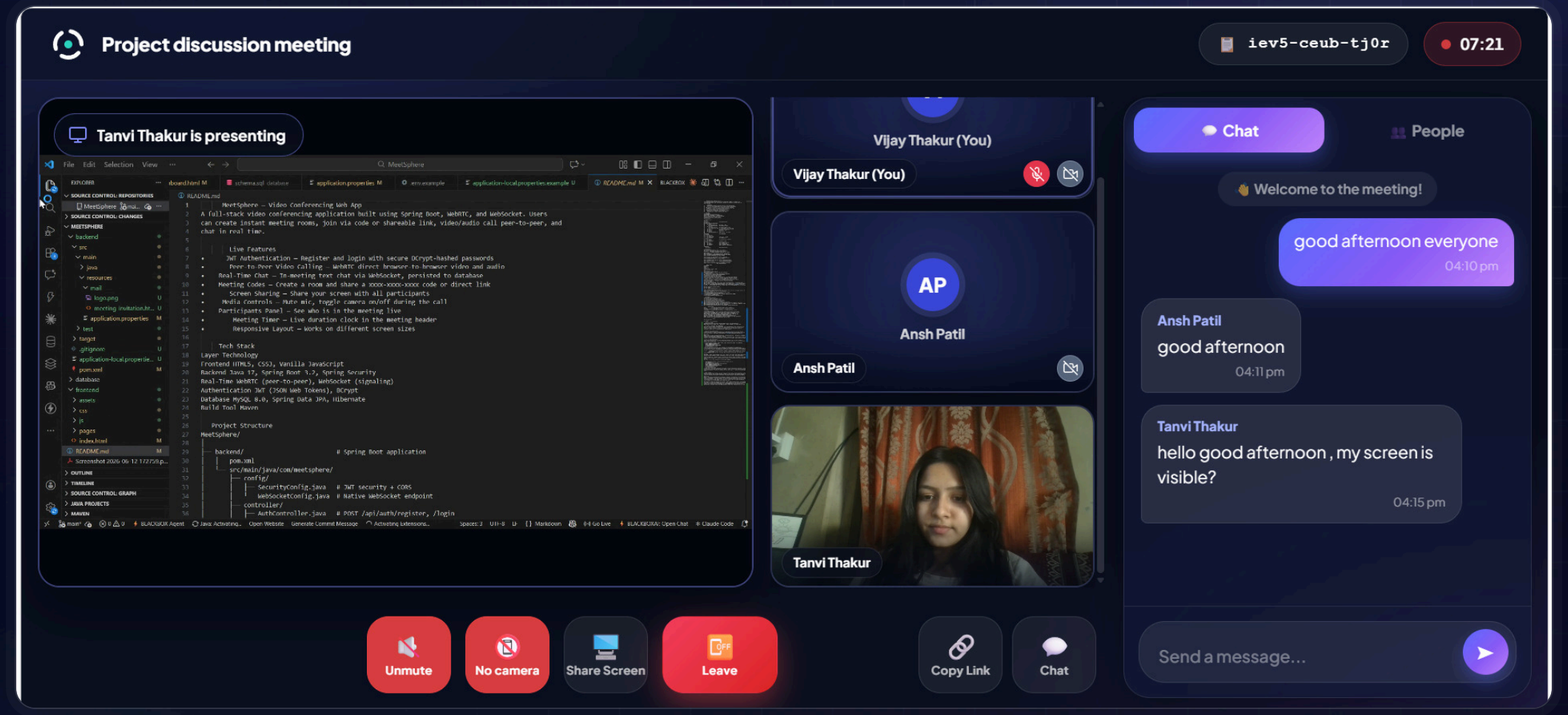
Viewing a Shared Screen



When a participant shares their screen, other participants can view the presentation in an enlarged area while participant tiles remain visible alongside it.

Live Messages | Sender Names | Timestamps | Chat Alongside Screen Sharing

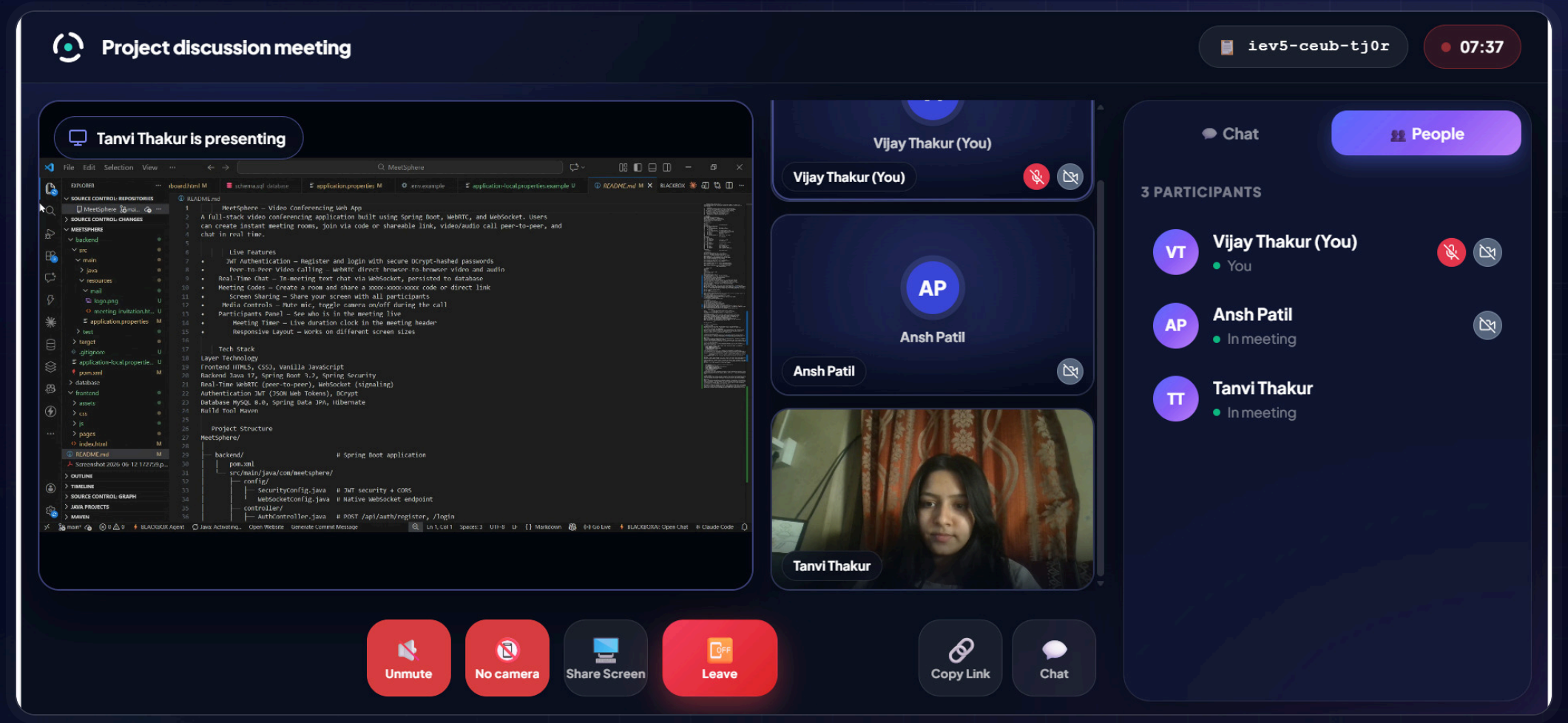
In-Meeting Chat



MeetSphere includes an in-meeting chat interface that allows participants to exchange messages while remaining in the meeting. Messages display sender information and timestamps.

Participant List | Microphone Status | Camera Status

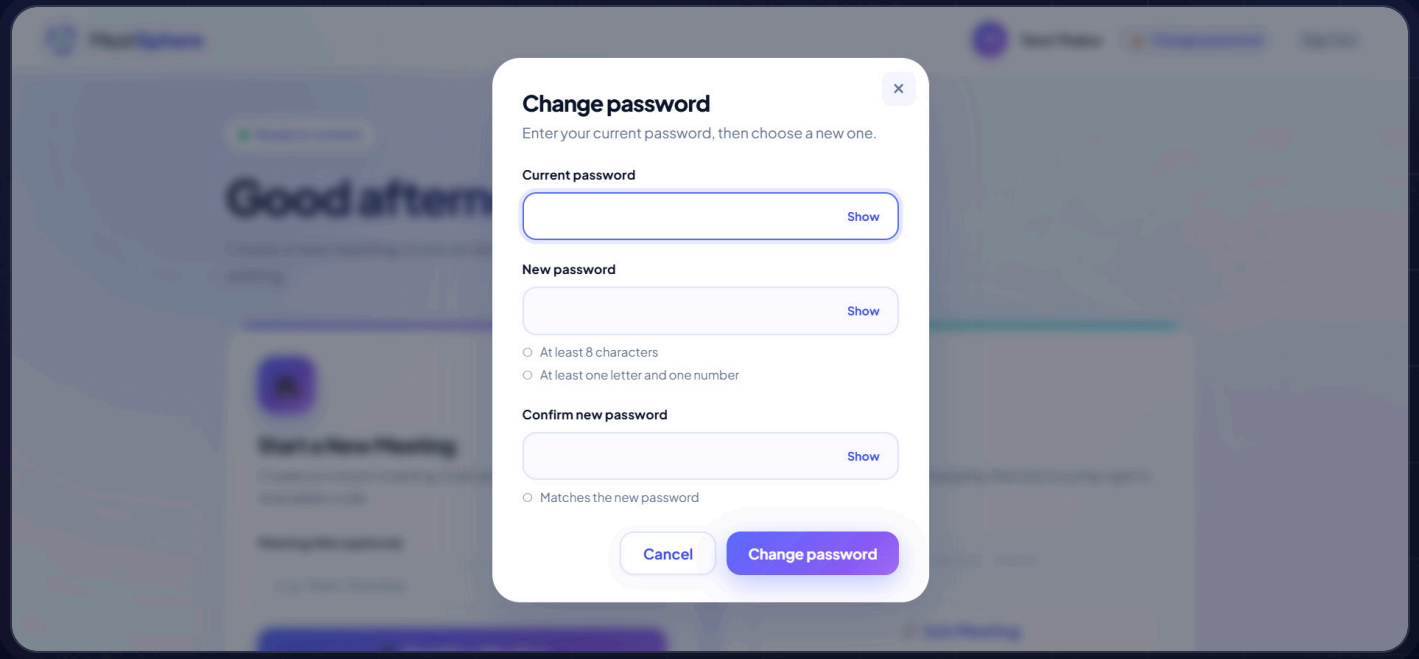
Live Participant Status



The participants panel displays the people in the meeting and provides visual indicators for microphone and camera states.

07 AUTHENTICATION & EMAIL

Authentication and Email Integration



Password management. The change-password interface allows users to enter their current password, a new password, and password confirmation. The interface displays password requirements.

JWT | BCrypt | Spring Security | Spring Boot SMTP

Authentication

JWT-based authentication is used to manage user sign-in. According to the project documentation, passwords are stored as BCrypt hashes.

Authorization

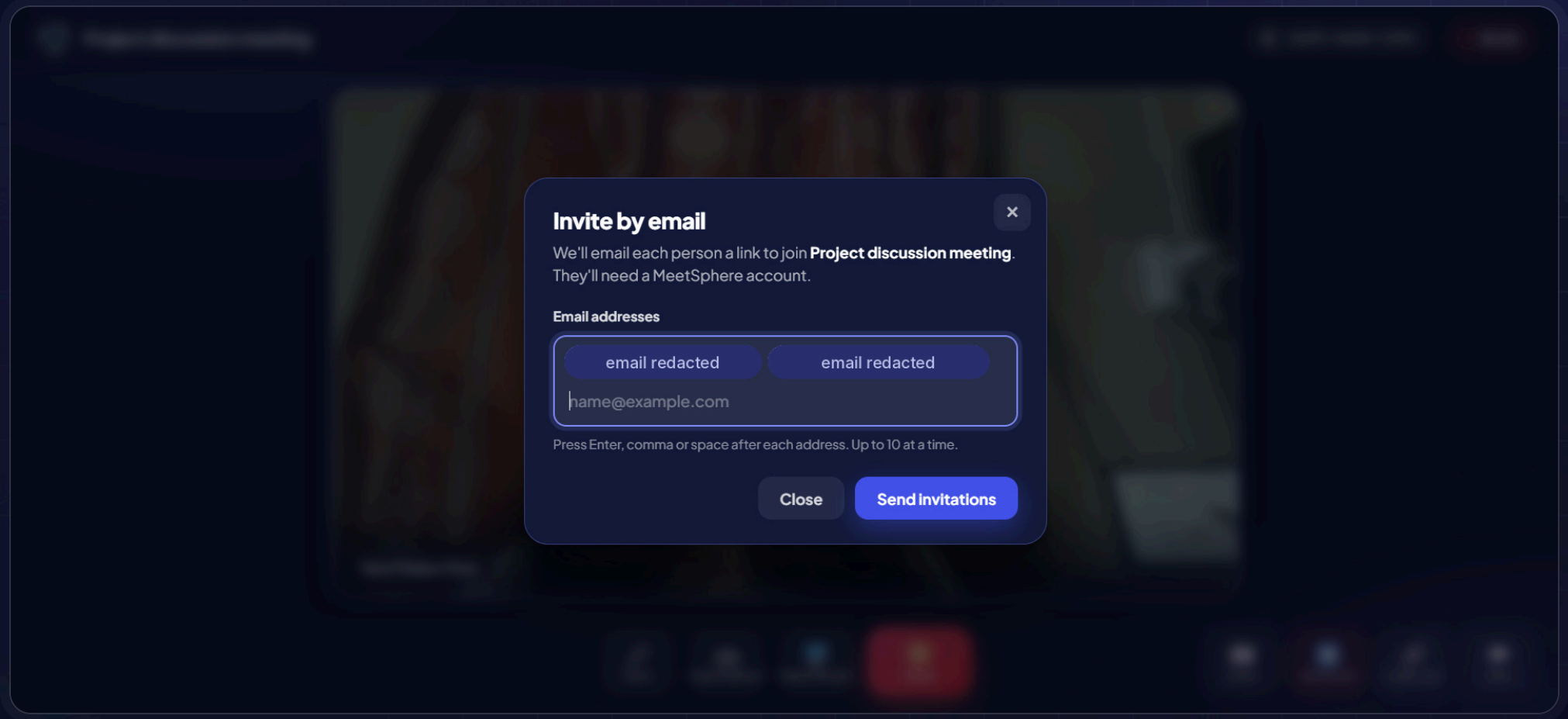
Spring Security is used to protect backend functionality and control access to secured resources.

Email Integration

Spring Boot SMTP integration supports sending meeting invitation emails to participants.

Multiple Recipients | Meeting Invitation | Privacy-Protected Email Addresses

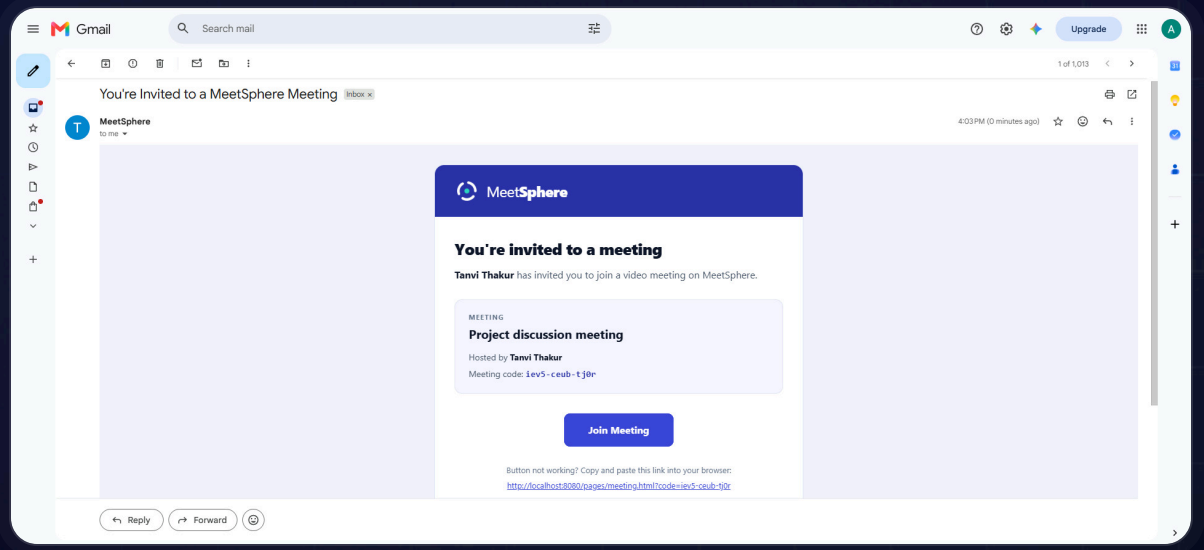
Inviting Participants by Email



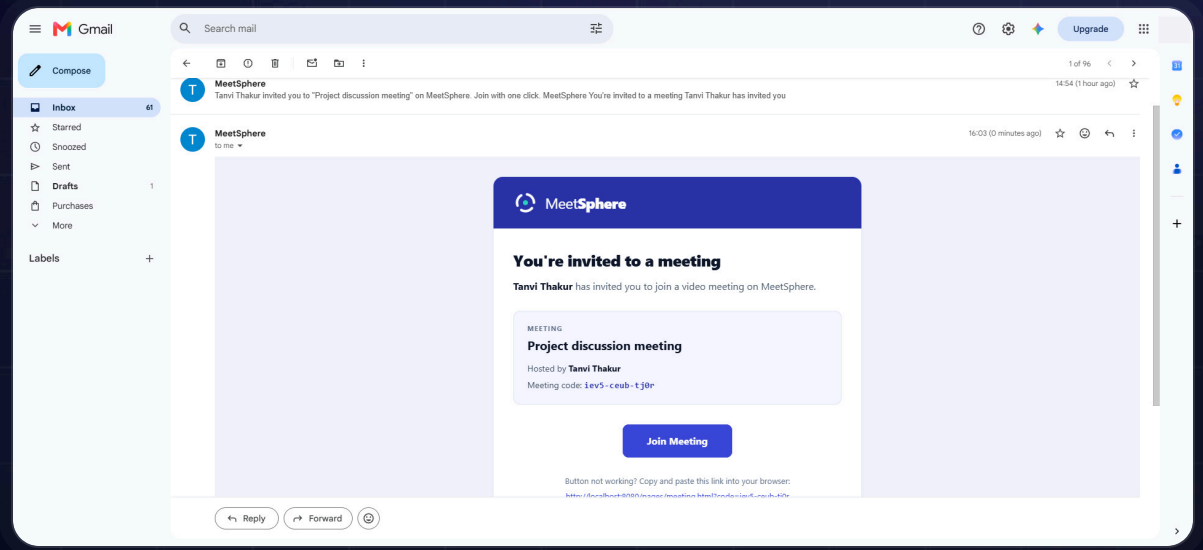
The meeting host can enter multiple email addresses and send meeting invitations containing a joining link. According to the current application workflow, invitees need a MeetSphere account to join.

07 AUTHENTICATION & EMAIL

Meeting Invitation Email



Inbox view. The invitation email displays the meeting title, host information, meeting code, and a button to join the meeting.



Thread view. The same invitation email shown in a conversation view.

BRANDED EMAIL

An HTML email template using MeetSphere branding.

MEETING DETAILS

The invitation includes relevant meeting information.

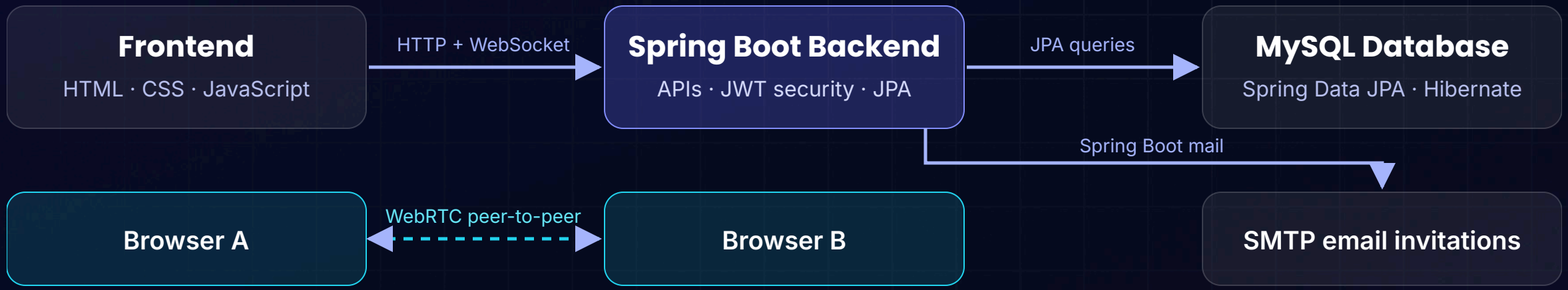
JOINING LINK

The email contains a link intended to open the meeting.

DEPLOYMENT NOTE The current joining link points to localhost because the application is running locally. The application is not publicly deployed.

Technology Stack and Architecture

FRONTEND	BACKEND	DATABASE	AUTHENTICATION	REAL-TIME COMMUNICATION	BUILD AND EMAIL
HTML5 CSS3 JavaScript	Java 17 Spring Boot 3.2 Spring Security	MySQL 8.0 Spring Data JPA Hibernate	JWT BCrypt	WebRTC WebSocket	Maven Spring Boot SMTP



The frontend communicates with the Spring Boot backend through HTTP APIs and WebSocket connections. The backend handles application logic, authentication, and database operations through Spring Data JPA and Hibernate.

WebRTC supports peer-to-peer audio, video, and screen-sharing functionality. WebSocket is used for signaling and real-time messaging as implemented in the application. Spring Boot SMTP integration is used to send email invitations.

09 KEY FEATURES & PROJECT LEARNINGS

Features and Project Learnings

Key Features Demonstrated

- ✓ Account registration, sign-in, and password management.
- ✓ Dashboard for creating or joining meetings.
- ✓ Meeting rooms with meeting codes and a duration timer.
- ✓ Camera and microphone controls.
- ✓ Screen sharing with a presentation layout.
- ✓ Live in-meeting chat with an unread-message indicator.
- ✓ Participant panel with status indicators.
- ✓ Email invitations containing meeting details and a joining link.

What I Explored and Learned

01 Connecting frontend interfaces to backend APIs.

02 Developing backend functionality with Java and Spring Boot.

03 Database operations using MySQL, JPA, and Hibernate.

04 Authentication and authorization concepts.

05 Real-time communication using WebRTC.

06 Real-time messaging and signaling using WebSocket.

07 Email integration using Spring Boot SMTP.

08 Organizing a full-stack application into frontend, backend, and database components.

This project provided practical experience in full-stack application development, integrating multiple technologies, testing application workflows, and understanding real-time communication concepts.

10 PROJECT SUMMARY

MeetSphere at a Glance

MeetSphere is a personal full-stack video conferencing application that brings together frontend development, backend APIs, database integration, authentication, and real-time communication in one application.

Frontend	Backend	Database	Authentication and Communication	Build and Email
HTML, CSS, JavaScript	Java, Spring Boot, Spring Security	MySQL, Spring Data JPA, Hibernate	JWT, WebRTC, WebSocket	Maven, Spring Boot SMTP

A personal full-stack project focused on video conferencing, real-time communication, backend development, database integration, and application security.

The application currently runs locally and is not publicly deployed.